IMO 2026 — Problem 1

Solution by xAI Grok 4.5

2026-07-21

Source: `results/grok-4.5/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\mathrm{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

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Approaches tried

- Elementary potential on number of entries > 1 and total product (for termination) — works.
- p-adic valuations: the gcd of the multiset of exponents for each prime is invariant under moves; this determines M uniquely — works.
- Direct simulation of many instances confirms path-independence of M and the closed form $M = \prod_p p^{g_p}$.

Current best

Parts (a) and (b) are fully proved. The final M equals $\prod_p p^{g_p}$ where g_p is the gcd of the initial p-adic valuations of the 2026 numbers (an invariant of every move). Termination follows from the lexicographic potential (k, P), where k is the number of entries > 1 and P is the product of all board entries.

Full proof

We work throughout with an ordered N-tuple of positive integers on the blackboard, where $N = 2026$; initially every entry is an integer strictly greater than 1. For integers $m, n > 1$ write

$$g := \gcd(m, n), \qquad \ell := \frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}.$$

A legal move consists of selecting two distinct board positions whose current entries m and n both exceed 1, and replacing those two entries by g and ℓ (in either order). The elementary identities

$$mn = \gcd(m, n) \cdot \operatorname{lcm}(m, n) \qquad \text{and} \qquad g \cdot \ell = \operatorname{lcm}(m, n) = \frac{mn}{g} \tag{1}$$

will be used repeatedly. In particular the product of the two new entries equals mn/g. Since $m, n > 1$ we have $\operatorname{lcm}(m, n) \geq \max(m, n) > 1$, so $g \cdot \ell > 1$ and therefore at least one of g, ℓ is strictly greater than 1.

Preliminary facts on valuations

Lemma 1 (effect of a move on valuations).
Fix a prime p. Let $\alpha = v_p(m)$ and $\beta = v_p(n)$. Then

$$v_p(g) = \min(\alpha, \beta), \qquad v_p(\ell) = \max(\alpha, \beta) - \min(\alpha, \beta).$$

Consequently

$$\gcd(v_p(g), v_p(\ell)) = \gcd(\alpha, \beta).$$

Proof.
By the standard formulae for the valuations of the gcd and the lcm we have $v_p(g) = \min(\alpha, \beta)$ and $v_p(\operatorname{lcm}(m, n)) = \max(\alpha, \beta)$, whence

$$v_p(\ell) = \max(\alpha, \beta) - \min(\alpha, \beta).$$

Without loss of generality $\alpha \leq \beta$. The new pair of valuations is then $(\alpha, \beta - \alpha)$, and the identity $\gcd(\alpha, \beta - \alpha) = \gcd(\alpha, \beta)$ is elementary. The case $\beta \leq \alpha$ is symmetric. $\square$

Lemma 2 (invariance of the gcd of a multiset of exponents).
Let S be a finite multiset of non-negative integers. Suppose two elements $\alpha, \beta \in S$ are replaced by $\min(\alpha, \beta)$ and $|\alpha - \beta|$. Then the greatest common divisor of all elements of the multiset remains unchanged.

Proof.
Write $d = \gcd(S)$. Then d divides both α and β, so d divides both $\min(\alpha, \beta)$ and $|\alpha - \beta|$; hence d still divides every element of the new multiset. Conversely, any common divisor of the new

multiset divides both new numbers and therefore divides their sum (equal to $\max(\alpha, \beta)$) and their difference (equal to $|\alpha - \beta|$), and consequently divides both α and β. It therefore divides every element of the old multiset. □

Corollary 3.
For every prime p define

$$g_p := \gcd\{v_p(x) : x \text{ is currently written on the blackboard}\},$$

adopting the conventions $\gcd(\emptyset) = 0$ and $\gcd(0, a) = a$. Then every legal move leaves each g_p unchanged. In particular, each g_p equals the value computed from the initial board.

Proof.
A legal move replaces a pair of board entries m, n by g and ℓ. By Lemma 1 the corresponding pair of p-adic valuations is replaced by $\min(\alpha, \beta)$ and $|\alpha - \beta|$. Lemma 2 then asserts that the gcd of the full multiset of N valuations is unchanged. (Primes not dividing mn contribute the pair $(0, 0)$, which is replaced by $(0, 0)$.) □

Part (a): termination with exactly one entry greater than 1

Let k denote the number of board entries that are strictly greater than 1, and let P denote the ordinary product of all N entries currently on the board. Both quantities are positive integers at the outset ($k = N \geq 2$). We claim that every legal move strictly decreases the ordered pair (k, P) in the lexicographic order on $\mathbb{N} \times \mathbb{N}_{>0}$.

Consider a legal move performed on entries $m, n > 1$, replaced by g and ℓ. There are three exhaustive possibilities according to the number of replacement entries that exceed 1:

1. *Both* $g > 1$ and $\ell > 1$. Then k is unchanged. From (1) the new product equals the old product divided by g. We claim $g \geq 2$. Indeed, if $g = 1$ then $\ell = \operatorname{lcm}(m, n) = mn > 1$, so exactly one of the two replacement numbers would exceed 1, contradicting the standing assumption of this case. Hence $g \geq 2$ and the new product P' satisfies $P' = P/g \leq P/2 < P$. The pair (k, P) therefore decreases lexicographically.

2. *Exactly one* of g, ℓ exceeds 1. Then two numbers greater than 1 are removed and exactly one number greater than 1 is written, so the new value of k equals the old value minus 1. The first component of the potential drops, regardless of the behaviour of P.

3. *Neither* g nor ℓ exceeds 1. This is impossible: $g \cdot \ell = \operatorname{lcm}(m, n) > 1$, so at least one factor is greater than 1.

Thus every move strictly decreases (k, P) lexicographically. Since both components are positive integers (or non-negative for k), only finitely many successive decreases are possible. Consequently only finitely many moves can be performed.

When no further move is possible we necessarily have $k \leq 1$, for otherwise two entries greater than 1 would still be available. It remains to rule out the possibility $k = 0$ at termination. Observe that a move decreases k only in case 2 above, and in that case the decrease is by exactly 1. Starting from $k = N \geq 2$, after any finite sequence of moves we therefore still have $k \geq 1$. (Equivalently: $g \cdot \ell > 1$ always, so a move that begins with two entries greater than 1 never produces two ones; the count of entries greater than 1 drops by at most one and cannot jump from 2 to 0.)

Hence at termination we have exactly $k = 1$: precisely one integer M on the blackboard is greater than 1, and every other entry equals 1.

This completes the proof of part (a).

Part (b): uniqueness of M

By part (a), every sequence of legal moves terminates after finitely many steps with a board consisting of a single integer $M > 1$ together with $N - 1$ ones. For this terminal configuration and for every prime p,

$$g_p = \gcd(v_p(M), v_p(1), \dots, v_p(1)) = \gcd(v_p(M), 0, \dots, 0) = v_p(M).$$

By Corollary 3 the integers g_p are independent of the particular sequence of moves; they are completely determined by the initial board. Therefore, for every prime p, the p-adic valuation of the terminal number M is independent of the sequence of moves. Since a positive integer is uniquely determined by its complete set of p-adic valuations, the value of M itself does not depend on the choices of Confucius.

Explicitly, if $a_1, \dots, a_N$ denote the numbers initially written on the blackboard, then

$$M = \prod_p p^{g_p}, \qquad \text{where} \qquad g_p = \gcd(v_p(a_1), \dots, v_p(a_N)),$$

the product running over all primes (only finitely many exponents being positive).

This completes the proof of part (b).

Remarks

- The same argument applies verbatim to any finite number $N \geq 1$ of initial integers greater than 1 (when $N = 1$ there are no moves and M is already the unique entry).
- While the hypothesis that every initial entry exceeds 1 guarantees that moves are available at the outset whenever $N \geq 2$, the proofs of the invariance of the g_p and of the shape of the terminal M require only that the entries be positive integers.